

LAB - In class

Topic 9: Transaction & Isolation Levels



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Notes

Put **ALL** SQL statements to **ONE** text (SQL) document names student ID <StudentID>.sql and a pdf document answer the questions <StudentID>.pdf then put into a zip/rar <StudentID>.zip:

- Comment Student ID and Full name on top of the file
- Run ALL lines in file without any error

In-class assignment

Use the University database (run UniversityDB_script file on moodle to create the database). To do this assignment, you may open two separate query windows/connections by simply opening 2 script files in SSMS. (Noted as session 1 and session 2 in the scenarios)

- Write a Stored Procedure to Get Department Payroll Summary:
 - + Input: DepartmentID
 - + Output: department_name, NumberOfInstructors , TotalSalary.
- Write a Stored Procedure to Adjust Instructor Salary
 - + Input: InstructorID, NewSalary
 - + Output: Commit If the instructor **exists** else Rollback
- Write Stored Procedure to Add New Instructor
 - + Input: InstructorID, InstructorName, Phone, DepartmentID, Salary
 - + Output: Commit If the instructor **does not exist** else Rollback

Use transactions to ensure that the stored procedures work as required. If both transactions are executed simultaneously, is there a risk of concurrency conflicts? (Try to give each isolation level a scenario if possible). If so, explain why, and propose a solution to fix the issue in the pdf document.

Note: You are prohibited from using Select with locks in this assignment.